Advanced Algorithms
COMP4121

Aleks Ignjatović

School of Computer Science and Engineering
University of New South Wales

Voting and Customer Feedback Aggregation

Voting and Customer Feedback Aggregation

(More details and more information can be found in Lecture Notes on voting, available on the course website)

- Can we trust customer feedback?
- Customer feedback aggregation suffers from a huge amount of fake feedback.
- How can we handle such a problem? Customer feedback is indispensable in making decisions what to buy, for example.
- **Bayesian aggregation:** (I never understood what is "Bayesian" in the method described below)
  - Select all products that are in a direct competition as available choices.
    - For example, all TVs with exactly the same screen size and the same features, but by different manufacturers.
    - Another example is all versions of the same product, for example, all models of MacBook Pro.
    - Roughly speaking, that would be the options the customers get to choose from when they search for a product.

Bayesian aggregation

- Bayesian aggregation (continued):
  - Let us consider a product i and assume that there are K products (including product i) in the same category, i.e., in a direct competition with product i ;
  - Assume that all of these K products have received N ratings in total.
  - Let the mean of all N ratings of these K products be R.
  - Assume also that there are n_i ratings of product i.
  - Let the mean of all ratings of product i be equal to r_i.
  - Then Bayesian aggregate rating of product i (which is what is reported to customers) is given by the formula

$$\rho_i = \frac{N \cdot R + n_i \cdot r_i}{N + n_i} = \frac{N}{N + n_i} R + \frac{n_i}{N + n_i} r_i$$

  - What is the idea behind this approach?

Bayesian aggregation

- Bayesian aggregation (continued):
  - If there are very few ratings of a product compared to the total number of ratings N of all products in the narrow category where that product belongs, these ratings cannot be trusted much.
  - They are likely to have large variance around the mean; in a sense the mean "has not stabilised" yet to be reasonably trusted.
  - To be on the safe side, we report (almost) the mean R of all similar products.
  - As the number of ratings n_i of product i increases, the mean r_i becomes more credible.
  - Thus, we report a weighted average of the "global mean" R of all very similar options and the actual mean r_i of ratings of product i.
  - The weights applied reflect the relative number of ratings of product i.

Bayesian aggregation

- Bayesian aggregation (continued):
  - If N is much larger than n_i, even when n_i is large, the impact of the mean r_i of ratings of product i on the Bayesian rating ρ_i is diminished and the dynamic range of all ratings is reduced too much.
  - For that reason some systems limit N by using instead

$$\begin{aligned}\rho_i &= \frac{\min(N, N_0) \cdot R + n_i \cdot r_i}{\min(N, N_0) + n_i} \\ &= \frac{\min(N, N_0)}{\min(N, N_0) + n_i} \cdot R + \frac{n_i}{\min(N, N_0) + n_i} \cdot r_i\end{aligned}$$

    where N_0 is the upper limit on how much N is allowed to grow.
  - A well known US beer review website "The Beer Advocate" uses such a formula, so it must be working well, given the significance of the domain.

Adaptive Vote Aggregation

- **Setup:** We will assume that a set of N *voters* $V_1, \ldots, V_N$ are given a collection of L lists $\Lambda_1, \ldots, \Lambda_L$.
- Each list Λ_l contains n_l many candidates $\Lambda_l = \{I_1^l, \ldots, I_{n_l}^l\}$, and the voters are asked to chose the "best" candidate on each list.
- Not every voter is obliged to vote for the best candidate on every list, but can choose to vote on a subset of these lists.

- **Problem:** As the system receives these votes, the task is to assess:
  1. the level of "community approval" for each candidate on every list.
  2. the reliability or *trustworthiness* of each voter, i.e., the degree to which his choices agree with the sentiment of the community.

Adaptive Vote Aggregation

- In order to make such estimate of the level of "community approval" robust against possible unfair voting practices of the participants, the assessment of the trustworthiness of voters should have the following features:
    1. Voters who vote on a large number of lists, and whose choices are largely in agreement with the prevailing sentiment of the community of voters, should obtain a higher level of trustworthiness than the voters who vote on only a few of these lists, or vote incompatibly with the prevailing sentiment.
    2. The impact on the final outcome of each voter should be proportional to (or positively correlated with) his trustworthiness.
    3. Voters who seldom vote or voters who vote more or less randomly, just to be seen as active in the community, and then choose to vote unfairly on a few particular lists for candidates which are favoured by few other voters, should not be able to secure election of their choices even if such colluding voters are a majority for those particular lists.

Adaptive Vote Aggregation

- Here "voting" might be a real voting, say for city councillors, such as the Lord Mayor, the Treasurer and other city officials.
- For each position there is a list of candidates who are running for such a position and voters are asked to pick their favourite candidate.
- However, many other problems are reducible to a voting procedure.
- For example, if customers are asked to rate a product i, this can be interpreted as voters (customers giving feedback) being asked to chose their favourite candidate on a list associated with product i, where candidates are different numbers of stars to be awarded to that product.

An Iterative Vote Aggregation Algorithm

- $r \to li$ denotes the fact that voter V_r has participated in voting for the best candidate on list Λ_l and has chosen candidate I_i^l;
- n_l denotes the number of candidates on a list l; thus, $n_l = |\Lambda_l|$;
- For each candidate I_i^l on list Λ_l we will keep track of its *raw level of community approval* at the step of iteration k, denoted by $r_{li}^{(k)}$ as well as of its *normalised level of community approval* at the step of iteration k denoted by $\rho_{li}^{(k)}$.
- For each voter V_r we will also keep track of his *trustworthiness* $T_r^{(k)}$ at the stage of iteration k.
- As before, we denote by $\|\boldsymbol{x}\|$ the usual length (the norm) of the vector $\boldsymbol{x} = \langle x_1, \ldots, x_n \rangle$, i.e., $\|\boldsymbol{x}\| = \sqrt{\sum_{i=1}^{n} |x_i|^2}$.

An Iterative Vote Aggregation Algorithm

An Iterative Vote Aggregation Algorithm

Initialization: Let $\varepsilon > 0$ be the precision threshold, $\alpha \geq 1$ a discrimination setting parameter and

$$\begin{aligned} T_i^{(0)} &= 1; \\ r_{li}^{(0)} &= |\{r \,:\, r \to li\}|; \\ \rho_{li}^{(0)} &= \frac{r_{li}^{(0)}}{\sqrt{\sum_{1 \leq j \leq n_l} (r_{lj}^{(0)})^2}}. \end{aligned}$$

Repeat:

$$\begin{aligned} T_r^{(k+1)} &= \sum_{l,i\,:\,r \to li} \rho_{li}^{(k)}; \\ r_{li}^{(k+1)} &= \sum_{r\,:\,r \to li} \left(T_r^{(k+1)}\right)^{\alpha}; \\ \rho_{li}^{(k+1)} &= \frac{r_{li}^{(k+1)}}{\sqrt{\sum_{1 \leq j \leq n_l} (r_{lj}^{(k+1)})^2}}; \end{aligned}$$

until: $\|\boldsymbol{\rho}^{\,(k+1)} - \boldsymbol{\rho}^{(k)}\| < \varepsilon$.